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The Ultimate Guide to Deploy LLaVA 1.6 Locally: A Journey Through Dependencies and Discoveries 🚀

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What I learned after spending way too much time making a multimodal AI actually work

Introductions Why Another LLaVA Guide?

You know that feeling when you follow a tutorial and somehow end up with more errors than when you started? Yeah, me too.

After wrestling with LLaVA 1.6 GUI deployment on an old-school HPC system(with the delightfully vintage GLIBC 2.28), I've collected a treasure trove of "interesting" moments.

This guide is essentially my notes to future me, but hopefully it saves you from the same scenic route through dependency hell.

What You'll Actually Get Working

By the end of this guide, you'll have:

- LLaVA 1.6 GUI running smoothly with ~12GB GPU memory usage
- 4-second model loading (after the initial download torture)
- Production-ready inference that actually works
- Solutions to every common error you'll encounter

Tested Environment: Linux RHEL 8, NVIDIA L40S, GLIBC 2.28 (Works for any GPU with 16GB+ VRAM for single inference and system overhead, but 24GB+ is recommended for easy handling multiple concurrent requests)

The Pre-Flight Reality Check

Before we embark on this adventure, let's check our supplies:

```
# Quick system check - save yourself time later
nvidia-smi --query-gpu=name,memory.total --format=csv
df -h /home --output=avail | tail -1 # Need 50GB+ free
ldd --version | head -1 # Check your GLIBC version
python --version # Will be useful in a bit
```

Friendly reminder: If you're running low on disk space, now's a great time to find a bigger drive. Trust me, running out of space mid-download is about as fun as 8 wisdom.

Step 1: Environment Setup (The Foundation)

Here's where things get interesting — most tutorials assume conda and pip are best friends. Spoiler alert: they have a complicated relationship.

```
# Create the environment
conda create -n llava_16 python=3.10 -y
conda activate llava_16

# Install torch with the right wheel for your CUDA version
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118

# Verify installation
python -c "import torch; print(f'Torch: {torch.__version__}, CUDA: {torch.cuda.is_available()}')"
```

Expected output: `Torch: 2.3.1+cu118, CUDA: True`

Step 2: Dependencies Without the Drama

```
# Install core dependencies
pip install transformers accelerate bitsandbytes sentencepiece protobuf

# Clone the repo and install requirements
git clone https://github.com/haotian-liu/LLaVA.git
cd LLaVA
pip install -e .
pip install gradio

# Test the installation
python -c "from llava.model.builder import load_pretrained_model; print('Success!')"
```

Step 3: Authentication

Time for the administrative side of things. Get your HuggingFace token:

1. https://huggingface.co/settings/tokens
2. Click "New token" → "Read" permissions
3. Copy that `hf_` token (it's like a backstage pass)

```
# The straightforward approach
export HUGGINGFACE_TOKEN="hf_your_token_here"
huggingface-cli login

# Save your token when prompted
# It will ask: "Token:" - paste your token
# It will ask: "Add token as git credential?" - type 'Y'

# Verify it worked
huggingface-cli whoami
# Should show your username and your token permissions

# Set for HF_HUB specifically (some libraries need this)
export HF_TOKEN=$HUGGINGFACE_TOKEN

# Make it stick across sessions
echo "export HF_TOKEN=\$HUGGINGFACE_TOKEN" >> ~/.bashrc
```

Step 4: Storage Strategy

Here's something I discovered: the model download is 13GB. Your default cache location (usually `~/.cache`) may be your HPC system/quota/home... options abound.

```
# Check if your cache location is feeling generous
df -h ~/.cache

# If space is tight, let's find a more spacious home
# Common alternatives: /scratch, /tmp, /data, /opt
mkdir -p /tmp_ssd/$USER

# Make it stick around (optional but recommended)
echo "export HF_HOME=/tmp_ssd/$USER/huggingface_cache" >> ~/.bashrc
```

Step 5: The Model Loading

Here's the battle-tested code that handles all the edge cases:

```
# Create a test script to verify everything works
cat > test_llava.py << EOF
import torch
from llava.model.builder import load_pretrained_model
from llava.mm_utils import get_model_name_from_path

# Model configuration - adjust these if you have different setup
model_path = "liuhaotian/llava-v1.6-vicuna-7b"
model_name = get_model_name_from_path(model_path)

# Load the model - this will take a while the first time
tokenizer, model, image_processor, context_len = load_pretrained_model(
    model_path=model_path,
    model_base=None,
    model_name=model_name,
    load_8bit=False,
    load_4bit=False,
    device="cuda"
)

print("✅ Model loaded successfully!")
print(f"Context length: {context_len}")
print(f"Vocabulary size: {len(tokenizer)}")
EOF

# Run it and watch the magic (or errors)
python test_llava.py
```

It might take 2 minutes for the first load, but a few seconds every time after!

The Error Museum (And Their Solutions)

Exhibit A: The Flash Attention Mystery

```
ImportError: cannot import name 'LlamaFlashAttention2' from 'transformers.models.llama.modeling_llama'
```

Translation: Your GLIBC is feeling vintage compared to flash attention's expectations.
Solution: Embrace eager attention (it's maybe 5% slower, but which counting?)
Good news: Our loading code handles this gracefully.

Exhibit B: The Great Space Shortage

```
RuntimeError: [Errno 28] No space left on device: '/home/.cache/huggingface/hub/models--liuhaotian--llava-v1.6-vicuna-7b/tmp0xyz123'
```

Translation: That 25GB download just made friends with your disk capacity
Solution: Point HF_HOME somewhere more spacious
Prevention: The `df -h` command is your friend

Exhibit C: The PyTorch Import Enigma

```
ImportError: /lib64/libc.so.6: version `GLIBC_2.29' not found (required by /home/.local/lib/python3.10/site-packages/torch/lib/libtorch_cpu.so)
```

Translation: Conda and pip had a disagreement about PyTorch
Solution: Clean state approach

```
# Nuclear option - start fresh
conda deactivate
conda remove -n llava_16 --all -y
conda create -n llava_16 python=3.10 -y
conda activate llava_16

# Install torch first, from the right source
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118

# Verify it's happy
python -c "import torch; print(torch.__version__)"
```

Exhibit D: The Authentication Time Loop

```
OSerror: We couldn't connect to 'https://huggingface.co' to load this file, couldn't find it in the cached files and it looks like liuhaotian/llava-v1.6-vicuna-7b is not the path to a directory containing a file named config.json.
Checkout your internet connection or see how to run the library in offline mode at https://huggingface.co/docs/transformers/installation#offline-mode.
```

Translation: Cached credentials gone rogue
Solution: Fresh start:

```
# Clear the HF cache
rm -rf ~/.cache/huggingface/token
huggingface-cli logout
huggingface-cli login
```

Step 6: Test Drive Your Setup

Create this simple test script:

```
# Create a minimal test to verify everything works
cat > quick_test.py << EOF
import torch
from llava.model.builder import load_pretrained_model
from llava.mm_utils import get_model_name_from_path
from llava.eval.run_llava import eval_model

# Quick test - should load much faster now
print("Testing model loading...")
tokenizer, model, image_processor, context_len = load_pretrained_model(
    model_path="liuhaotian/llava-v1.6-vicuna-7b",
    model_base=None,
    model_name="llava-v1.6-vicuna-7b",
    load_8bit=False,
    load_4bit=False,
    device="cuda"
)

print("✅ Success! Model loaded in seconds!")
print(f"Memory usage: {torch.cuda.memory_allocated()/1024**3:.1f}GB")
EOF

# Run it
python quick_test.py
```

Step 7: Production Considerations

```
# Monitor GPU usage while running
watch -n 1 nvidia-smi

# Check memory usage programmatically
python -c "import torch; print(f'Allocated: {torch.cuda.memory_allocated()/1024**3:.1f}GB')"

# Clear GPU memory if needed
python -c "import torch; torch.cuda.empty_cache()"
```

The Pro Tips I Wish I Had

1. **Bookmark this exact version** - Dependencies change, what works today might break tomorrow
2. **Document your modifications** - Future you will forget what you changed
3. **Keep backup environments** - Conda environments can be exported and shared
4. **Monitor disk usage** - Models cache everywhere, clean periodically
5. **Test after updates** - Small changes can cascade into big problems

Performance Reality Check

Here's what success looks like:

```
# Expected performance metrics
time python quick_test.py
# Should load in 30-60 seconds vs 2+ minutes first time

# Memory usage should be reasonable
nvidia-smi --query-gpu=memory.used --format=csv
# Expect ~12GB for 7B model with quantization

# Check model cache size
du -sh ~/.cache/huggingface/hub/models--liuhaotian--llava-v1.6-vicuna-7b
# Should show ~13GB
```

Quick Troubleshooting Guide

```
# Common fixes in one place
# Space issues
rm -rf ~/.cache/huggingface/hub/models--*/tmp*
# Auth issues
huggingface-cli logout && huggingface-cli login
# Import issues
pip uninstall torch torchvision torchaudio -y && pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118
# Memory issues
export CUDA_VISIBLE_DEVICES=0
# Slow loading
export HF_HUB_ENABLE_HF_TRANSFER=1
```

Conclusion: The Journey Continues

Now you have a working LLaVA setup. The journey through dependency hell was worthwhile because:

- Your setup is reproducible
- You understand the common failure points
- You have multiple workarounds for different scenarios
- You can troubleshoot when things break

Next steps might include:

- Building a proper API wrapper
- Creating a batch processing script
- Integrating with your existing workflows
- Exploring other multimodal models

Happy modeling!

• • •

Software engineer by day, trail navigator by weekend. Writing about AI, class code, and how debugging works better with fresh air and no black notifications.

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